PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	General practitioners' attitudes towards opioids for non-cancer pain: a qualitative systematic review
AUTHORS	Punwasi, Rani; de Kleijn, L.; Rijkels-Otters, J.B.M.; Veen, M.; Chiarotto, Alessandro; Koes, Bart

VERSION 1 – REVIEW

REVIEWER	Wilson, Hester University of New South Wales
REVIEW RETURNED	02-Aug-2021

GENERAL COMMENTS	Thank you so much for undertaking this important synthesis of qual papers on GP experience of opioids and non-cancer pain. I understand that English is a second language for you and while it is very good, you need to ask an experienced English editor to review your manuscript, many sentences are weirdly constructed or have grammatical mistakes that interfere with the story you're telling. I think some of what is in the discussion would be better placed in the introduction. And some in the results. I'd also like to hear the researcher's voice in this qual review. Consider 'I' 'we.' You are reading and re-interpreting the papers, allow your interpretation to come through, you have a real strength in the fact that you are a GP. You are making sense of previous research through your lens as a GP. I've offered some more detailed comments.

REVIEWER	Steinke, Douglas The University of Manchester, Centre for Pharmacoepidemiology and Drug Safety
REVIEW RETURNED	09-Aug-2021

GENERAL COMMENTS	This is a well written and interesting article about the topic of opioids in chronic non-cancer pain; GPs perspective in a systematic review of qualitative research. It is interesting to bring together in a review the opinions and perspectives of GPs on this topic and found it interesting what the review had identified. It is useful information that should be used alongside quantitative research on opioid drug use to explain some of the issues. There are a couple of minor issues that are easily fixed that should be considered by the authors before publication: - page 16 under Gp's sense of pwerlessness - first sentence - please change don't to do not.

	- Figure 1, the flow diagram. Could you include at the first arrow downward, how many articles were removed because they were duplicates and further down, how many articles were removed before full text review (4 papers). Also on the left hand side, the first box should be expanded so it reads "Identification" (last n missing)

VERSION 1 – AUTHOR RESPONSE

We appreciate you and the reviewers for your time in reviewing our paper and providing valuable comments. We have carefully considered the comments and tried our best to address every one of them. We welcome further constructive comments if any. Please find attachted our point-by-point response.

We sincerely hope that the review team and editors of BMJ open agree with the drastic revisions that we have made after reviewing the comments of the reviewers. And we hope that this rewritten manuscript is accepted for publication in your journal. We look forward to hearing from you in due time and welcome any further questions and comments you may have.

VERSION 2 – REVIEW

REVIEWER	Wilson, Hester University of New South Wales
REVIEW RETURNED	18-Oct-2021

GENERAL COMMENTS	Thank you for allowing me to review this paper again. It's an important issue and great to see GPs doing this work. Please don't be disheartened, however I think your paper needs more work. There are still some grammatical issues, sentences that ae incomplete or don't entirely make sense in English and some English words that don't make sense. I also think your language is a little too colloquial and emotive. I also wonder whether you have adequately met the aims of your paper. What does your paper add to the work already done? 'The opioid crisis has gained a strong foothold in high-income countries'. I'm not sure what you mean by opioid crisis, I know this term is used a lot but what does it actually mean? If you use it here you need to use and explain it in your introduction. It sounds too emotive and nonspecific. What is a strong foothold? Need to be more specific. what is happening and why is this a problem? I know 'crisis' is used in the deseaux paper, but not sure this means you should also use this, I think for academic writing you need to be more specific and focused

	Worrisome – what do you mean why worrisome? 'fight the opioid crisis.' Once again I don't think this is great academic language. Its ok to use this in commentaries, or general journalism, but it pretty imprecise and I keep wondering what it is you are actually describing. What do you mean by chronic pain? What do you mean by attitude? This group issued a warning on opioid medicines to inform patients about the risk of addiction. Was it a warning, I think this is too emotive, perhaps recommendation, Is this what the group said, did they use the word 'addiction'? Despite these modest positive signs, the absolute numbers of opioid prescriptions in the Netherlands are still alarming. I think need to rethink use of word alarming, what are the problems? 'This is the first review on this topic which is conducted by members working in the field of primary care' Not sure this is true. Got to be careful with this….. and if it is true why is it important? Kennedy et al (2019) 'If you can't see a dilemma in this situation, you should probably regard it as a warning': a meta-synthesis and theoretical modelling of general practitioners' opioid prescription experiences in primary care. Suggest quality with to our knowledge Maybe say by GPs working in GP 'Worldwide a disturbing trend in doctor-prescribed opioid use is visible.' Why disturbing, and how? That's a bit emotive. Explain why you are saying this, what does this mean? Suggest 'worldwide we are seeing a trend in increased opioid prescribing.' 'the refill' sounds a bit colloquial may need to use more specific term 'Additionally, their addictive nature has led to the worrisome worldwide opioid crisis.' What do you mean by addictive nature and worrisome and crisis – need to explain why it is a crisis – what does that mean? Maybe use prescription opioid use disorder, rather than non specific addictive. These words are too emotive. 'Worldwide hospital admissions related to opioid use increased in past years.' Needs 'have' The opioid crisis in USA is not comparable to the crisis seen in Europe. You're assuming the reader knows about the opioid issues in US. Need to briefly describe this and how it is different Yet in Europe, too, the opioid crisis is alarming with increasing opioid prescription rates, opioid related hospitalizations and deaths. (15-18) It is of utmost importance to turn the tide worldwide. ' Get rid of 'alarming, '

	‘turn the tide’, what do you mean, what outcomes are you looking for? use instead decrease the numbers of opioid prescriptions for the treatment of chronic pain. Recent numbers provided by the Foundation for Pharmaceutical Statistics, an institute collecting prescription rates in the Netherlands, already demonstrate a decrease of opioid prescriptions after years of growth change to ‘has demonstrated’ the absolute numbers of – absolute number Netherlands are still alarming. Is rather than are, why alarming, please explain and support this with data. ‘Since our review team mainly consists of GP’s, we believe our review will generate a deeper level of understanding which may initiate practical changes in our everyday work that can address this crisis. Therefore, the aim of this study is to gain insight into barriers and facilitators influencing GPs’ opioid prescribing and to identify possibilities for improvement of current guidelines in order to promote opioid reduction in primary care.’ Are you wanting to promote opioid reduction? is this the aim? or is it about harm and better outcomes, I’m not sure that opioid reduction alone is the target. This is a review, its not original research. Surely the aim is to review and critique or analyse the current literature to gain insight? You’re making a virtue out of your positionality, inside status, but this has risks as well. ‘This study is reported following’ – this study followed ‘These GPs pledge for more public awareness and education among patients regarding their pain.’ Sorry I know you chose not to change this but this doesn’t make sense to me. But ‘pledge for more ‘doesn’t make much sense for me as an English speaker, its sound weird. I’d suggest instead, ‘plead’ would make more sense, or request, encourage, ‘pledge’ sounds odd to me. Is this a direct quote? If so, then needs to be in inverted commas. And the sentence needs to be rewritten as it doesn’t make sense. Because of the well-known addictive character of opioids, there is a stigma in prescribing opioids these days. I’m not sure this adequately describes stigma, are you saying that prescribing is stigmatised? if so how? What is important about stigma, how dos this play into the aims of your study? ‘order to alter the opioid crisis.’ Are they trying to alter the opioid crisis or are they trying to decrease potential harm for their patients. ‘history of substance misuse or patients with an “abusive” personality’ got to say I disagree with

	your use of personality here, this is about challenging behaviours not personality. This feels like stigmatising use of language and not helpful. Your theme 'GPs caught in the middle of the opioid crisis' is this language that the papers use or yours? I'm not sure the use of the word crisis is great, unless it is quoting directly from participants quotes. For osteoarthritis opioid treatment was seen as overtreatment by several GPs. This is not a complete sentence. GPs' experience plays a pivotal role in opioid prescription decision-making what do you mean by experience? I'm not sure that strong therapeutic relationships and years in practice are 'experience' are they expertise? GPs also reported to feel more confident in prescribing opioids needs rewriting, 'reported to feel more' doesn't make sense. In your discussion you state there are 4 themes but don't restate them. I think it makes sense to do this, remember the parts most likely to be read are the abstract and the first few lines of the discussion. The alarming numbers of opioid prescriptions and refill prescriptions among GPs – once again you need to explain this language As GPs we recognize the powerlessness felt when after hours of motivational talks with patients suffering chronic pain return from high quality pain centres with opioid prescriptions without further explanation or communication. This is a bit confusing??? Needs to be rewritten, non-cancer pain – you've changed your term…. This review describes how GPs' as "healers of pain" and "guardians of the community" deliberate the use of opioids in chronic pain. Doesn't make sneeze in English needs rewrite A zero-tolerance policy towards pain – not sure what you mean by this statement. effective communication strategies not only for GPs but also fine-tuned by GPs are necessary to turn the tide on this opioid crisis. This is new material and it's in your conclusion. It's an important issue and needs to be discussed above. You do mention communication with specialists above but this sentence is unclear do you mean with patients with staff with other Gps or with specialists? I think you're right and that the thing you have to offer that is different is he rethinking from a GP

	perspective, I'd love to read more about this, so expand on it if you can. Can you say that what would work for you in your setting is the some as GPs from other countries and other systems? If this is part of PhD you do need to mention that.

VERSION 2 – AUTHOR RESPONSE

Thank you for allowing me to review this paper again. It's an important issue and great to see GPs doing this work.

Please don't be disheartened, however I think your paper needs more work. There are still some grammatical issues, sentences that ae incomplete or don't entirely make sense in English and some English words that don't make sense.

Authors' response:

- It would be unprofessional to publish an article that has not been written in correct English. Hence, we appreciate your close look at our review. To accurately address these issues we have asked a native English speaker to read and edit our manuscript to tackle these grammatical issues.

I also think your language is a little too colloquial and emotive.

Authors' response:

- In addition we have asked the native speaker to help us reformulate these colloquial and emotive uses of words. Below, we will address all the comments point-by-point.

I also wonder whether you have adequately met the aims of your paper. What does your paper add to the work already done?

Authors' response:

- The aim of our study was to gain insight into barriers and facilitators influencing GPs' opioid prescribing and to identify possibilities for improvement of current guidelines in order to promote opioid reduction in primary care. This review has increased our understanding of why the decision to refrain from opioid prescription in pain management is complicated. Our findings highlight that, in order to promote opioid reduction in primary care and reduce the harms associated with opioid misuse, future research is needed to develop practical guidelines on appropriate opioid prescribing, tapering off opioid use and adopting effective communication strategies. In addition, this review points out that this research should be done in cooperation with GPs to adequately address the issues put forth by them in this qualitative review. We believe we have adequately met the aims. However, to address the concerns we have partly rewritten our conclusion to highlight the aims that were met in conducting this review, as follows (page 28, line 1-9): "This review demonstrates the difficulties encountered by GPs in treating (chronic) non-cancer pain and refraining from opioid prescription: a zero-tolerance policy towards pain by both doctors and patients; a wish for strong doctor-patient relationships with a fear of difficult conversations; a lack of knowledge and protocols on effective strategies to treat (chronic) pain in primary care; a lack of time; and inadequate collaboration with, and guidance from, specialists. Our findings highlight that in order to promote appropriate opioid prescription in primary care and to reduce the harms associated with opioid misuse, future research is needed to develop practical guidelines on appropriate opioid prescribing, tapering off opioid use and adopting effective communication strategies not only for GPs but also fine-tuned by GPs."

'The opioid crisis has gained a strong foothold in high-income countries'.

I'm not sure what you mean by opioid crisis, I know this term is used a lot but what does it actually mean? If you use it here you need to use and explain it in your introduction. It sounds too emotive and nonspecific. What is a strong foothold? Need to be more specific. what is happening and why is this a problem?

Authors' response:

- We have revised the first sentence of the abstract: "Worldwide the use of opioids, both doctor-prescribed and illicit, has increased." (page 2, line 4)
- To meet the concerns put forth by the reviewer, we have decided to replace the term "crisis" throughout the paper with more suitable terms.

I know 'crisis' is used in the deseaux paper, but not sure this means you should also use this, I think for academic writing you need to be more specific and focused

Authors' response:

- We decided to preserve the term “opioid crisis” in the introduction to introduce and explain this term as this term is often mentioned in the result section where it was cited from the included papers: “While this “opioid crisis”, as it is often called in the USA, is not comparable with the increase in opioid misuse in Europe, opioid prescription rates are nonetheless increasing and opioid-related hospitalizations and deaths are concerning.” (page 4, line 8-12)
- To meet the concerns put forth by the reviewer, we have decided to replace the term “crisis” throughout the rest of the paper with more suitable terms, such as “increased opioid prescribing” and “increase of opioid use and misuse”:
    - “Worldwide we are seeing a trend in increased opioid prescribing.” (page 3, line 26)
    - “In the UK, an Opioid Expert Working Group has been installed to address the increase of opioid use and misuse.” (page 4, line 16-18)

Worrisome – what do you mean why worrisome?

Authors’ response:

- We agree that using words such as “worrisome” are too emotive. We have therefore revised the sentence “Additionally, using opioids can lead to opioid tolerance, dependence and even addiction; it is partly this addictive nature of opioids that has led to an increase of prescription opioid use disorder. (page 4, line 3-5)

‘fight the opioid crisis.’ Once again I don’t think this is great academic language. Its ok to use this in commentaries, or general journalism, but it pretty imprecise and I keep wondering what it is you are actually describing.

Authors’ response:

- To meet the concerns, we have rewritten the sentence “In the UK an Opioid Expert Working Group has been installed to fight the opioid crisis” into “In the UK an Opioid Expert Working Group has been installed to address the increase of opioid use and misuse.” (page 4, line 16-18)

What do you mean by chronic pain?

Authors’ response:

- We thank the reviewer for pointing out that we did not define chronic pain in our review. We agree that using “chronic pain” in this review should be accompanied by a clear definition.

Hence, we have added the definition we used “pain lasting > 3 months (as defined by the NICE guidelines)” after the term chronic pain. (page 4, line 14-16)

What do you mean by attitude?

Authors’ response:

- By GP’s attitude we mean GP’s settled way of thinking or feeling.

This group issued a warning on opioid medicines to inform patients about the risk of addiction. Was it a warning, I think this is too emotive, perhaps recommendation, Is this what the group said, did they use the word ‘addiction’?

Authors’ response:

- Thank you for pointing out the emotive use of words in this sentence. We have rewritten the sentence using the word “recommendation” and to give a more general explanation of their work: “This group has come with multiple recommendations that should inform patients about the risk of opioid dependence and addiction.” (page 4, line 18-19)
- Indeed, the opioid expert working group issued a warning on opioid medicines to inform patients about the risk of addiction. Please see https://www.gov.uk/drug-safety-update/opioids-risk-of-dependence-and-addiction

Despite these modest positive signs, the absolute numbers of opioid prescriptions in the Netherlands are still alarming. I think need to rethink use of word alarming, what are the problems?

Authors’ response:

- Thank you for pointing out again the emotive use of words. We have rewritten the lines in order to clarify and specify the problem more adequately. “Despite these modest positive signs, the absolute numbers of opioid prescriptions in the Netherlands are still alarming. “ to “Despite these modest positive signs, more action is needed to further decrease opioid prescriptions numbers in coming years.” (page 4, line 24-26)

‘This is the first review on this topic which is conducted by members working in the field of primary care’ Not sure this is true. Got to be careful with this….. and if it is true why is it important?

Suggest quality with to our knowledge

Authors’ response:

- We appreciate the concerns put forth by the reviewer and decided to rewrite the sentence to “To the best of our knowledge, this is the first review on this topic conducted by professionals working directly in primary care.” (page 3, line 6-7)
- We think it is important to mention that our team consisted of mainly GP’s as in general practice most guidelines are generated by the discipline itself. To our best knowledge none of the previous publications on this topic have been conducted by primary care workers. Conducting a review on this topic by our team with our experience as GPs and our daily struggles with opioid prescriptions, we believe, will therefore generate a deeper level of understanding.

Maybe say by GPs working in GP

Authors’ response:

- Since not all authors are working as a GP, we have decided to rewrite the sentence to “To the best of our knowledge, this is the first review on this topic conducted by professionals working directly in primary care.” (page 3, line 6-7)

‘Worldwide a disturbing trend in doctor-prescribed opioid use is visible.’ Why disturbing, and how? That’s a bit emotive. Explain why you are saying this, what does this mean? Suggest ‘worldwide we are seeing a trend in increased opioid prescribing.’

- Thank you for your suggestion, we rewrote the sentence into “worldwide we are seeing a trend in increased opioid prescribing.’ (page 3, line 26)

‘the refill’ sounds a bit colloquial may need to use more specific term

Authors’ response:

- We replaced the term “the refill” by “repeat prescriptions”. (page 3, line 31)

‘Additionally, their addictive nature has led to the worrisome worldwide opioid crisis.’ What do you mean by addictive nature and worrisome and crisis – need to explain why it is a crisis – what does that mean? Maybe use prescription opioid use disorder, rather than non specific addictive. These words are too emotive.

Authors’ response:

- We rewrote the sentence in order to specify the addictive nature and added the suggestions. “Additionally, using opioids can lead to opioid tolerance, dependence and even addiction; it is partly this addictive nature of opioids that has led to an increase of prescription opioid use disorder.” (page 4, line 3-5)

‘Worldwide hospital admissions related to opioid use increased in past years.’ Needs ‘have’

Authors’ response:

- We have added the word “have” as was suggested by reviewer 1. (page 4, line 5-6)

The opioid crisis in USA is not comparable to the crisis seen in Europe. You’re assuming the reader knows about the opioid issues in US. Need to briefly describe this and how it is different

‘ Get rid of ‘alarming, ‘

Authors' response:

We agree with reviewer 1 and we have rewritten the lines to: "In the USA more than 4% of the adult population currently misuse prescription opioids, and the number of opioid-related deaths per year increased six-fold between 1999-2017. (15,16) While this "opioid crisis", as it is often called in the USA, is not comparable with the increase in opioid misuse in Europe, opioid prescription rates are nonetheless increasing and opioid-related hospitalizations and deaths are concerning. (17-20) It is of utmost importance to decrease inappropriate opioid prescription rates.'' (page 4, line 6-12)

'turn the tide', what do you mean, what outcomes are you looking for? use instead decrease the numbers of opioid prescriptions for the treatment of chronic pain.

Authors' response:

- We rephrased it into: "It is of utmost importance to decrease inappropriate opioid prescription rates." (page 4, line 11-12) We chose to not replace it by "decrease the numbers of opioid prescriptions for the treatment of chronic pain" as prescription rates should decrease not only for chronic pain but also in any other cases where more appropriate measures should be tried first.

Recent numbers provided by the Foundation for Pharmaceutical Statistics, an institute collecting prescription rates in the Netherlands, already demonstrate a decrease of opioid prescriptions after years of growth change to 'has demonstrated' the absolute numbers of – absolute number Netherlands are still alarming. Is rather than are, why alarming, please explain and support this with data.

Authors' response:

- We rephrased the sentence: "The Foundation for Pharmaceutical Statistics, an institute collecting prescription rates in the Netherlands, reported a 6% decrease in opioid prescriptions in 2019 compared to 2018, the first reduction seen after years of growth." (page 4, line 22-24)

'Since our review team mainly consists of GP's, we believe our review will generate a deeper level of understanding which may initiate practical changes in our everyday work that can address this crisis.

Therefore, the aim of this study is to gain insight into barriers and facilitators influencing GPs' opioid prescribing and to identify possibilities for improvement of current guidelines in order to promote opioid reduction in primary care.' Are you wanting to promote opioid reduction? is this the aim? or is it about harm and better outcomes, I'm not sure that opioid reduction alone is the target.

Authors' response:

- To clarify our aim we have rewritten the sentence to "Therefore, the aims of this study are to gain insight into GPs' attitudes and the barriers and facilitators influencing GPs' opioid prescription practices and to identify possible strategies to promote opioid reduction in primary care and to reduce the harm associated with opioid misuse." (page 5, line 1-4)

This is a review, its not original research. Surely the aim is to review and critique or analyse the current literature to gain insight?

Authors' response:

- Yes indeed, our aim is to review and analyse the current literature on this topic. And by reviewing primary studies we have tried to identify possible elements that can be used in opioid reduction strategies in primary care with the ultimate goal to reduce the harm associated with opioid misuse.

You're making a virtue out of your positionality, inside status, but this has risks as well.

Authors' response:

We are quite aware that our positionality might cause a source of bias. Yet in our understanding qualitative research always contains an element of subjectivity. Nonetheless, we believe that being involved in the field leads to a deeper level of understanding which may spur practical changes that are needed in our everyday work, and that these advantages outweigh the risk of bias due to subjectivity that is inherent to qualitative work.

'This study is reported following' – this study followed

Authors' response:

- We have rephrashed the sentence as was suggested by the reviewer.

These GPs pledge for more public awareness and education among patients regarding their pain.' Sorry I know you chose not to change this but this doesn't make sense to me. But 'pledge for more 'doesn't make much sense for me as an English speaker, its sound weird. I'd suggest instead, 'plead' would make more sense, or request, encourage, 'pledge' sounds odd to me. Is this a direct quote? If so, then needs to be in inverted commas. And the sentence needs to be rewritten as it doesn't make sense.

Authors' response:

- We replaced "pledge" by "pleaded" as suggested by the reviewer. (page 21, line 10)

Because of the well-known addictive character of opioids, there is a stigma in prescribing opioids these days. I'm not sure this adequately describes stigma, are you saying that prescribing is stigmatised? if so how? What is important about stigma, how dos this play into the aims of your study?

Authors' response:

- Some GPs reported prescribing opioids to have a stigma. We rephrased the sentence in order to clarify that this was a quote from a primary study: "Because of the well-known addictive character of opioids, some GPs reported a stigma in prescribing opioids". (page 21, line 13-14)

'order to alter the opioid crisis.' Are they trying to alter the opioid crisis or are they trying to decrease potential harm for their patients

Authors' response:

- We agree that this sentence required more clarification. We rephrased: "GPs felt caught between the desire to effectively treat pain and the societal obligation to decrease opioid prescriptions in order to reduce harm." (page 21, line 19-21)

'history of substance misuse or patients with an "abusive" personality' got to say I disagree with your use of personality here, this is about challenging behaviours not personality. This feels like stigmatising use of language and not helpful.

Authors' response:

- This was cited from Desveaux et al. We therefore have put "abusive" and also "personality" within quotation marks.

'GPs caught in the middle of the opioid crisis' is this language that the papers use or yours? I'm not sure the use of the word crisis is great, unless it is quoting directly from participants quotes.

Authors' response:

- In many of the cited articles the term "crisis" is used. Our thematic analyses showed GPs feel caught in the middle of this "crisis". Because this term was frequently used in the cited articles we have put the opioid crisis in the thematic themes between quotations mark.

For osteoarthritis opioid treatment was seen as overtreatment by several GPs. This is not a complete sentence.

Authors' response:

- We rephrased the sentence: "Opioid prescription was viewed as an overtreatment of osteoarthritis by several GPs."

GPs' experience plays a pivotal role in opioid prescription decision-making what do you mean by experience? I'm not sure that strong therapeutic relationships and years in practice are 'experience' are they expertise?

Authors' response:

- We replaced "experience" by "expertise" as suggested by the reviewer.

GPs also reported to feel more confident in prescribing opioids needs rewriting, 'reported to feel more' doesn't make sense.

Authors' response:

We rephrased the sentence: "GPs also reported increased confidence in opioid prescribing".

In your discussion you state there are 4 themes but don't restate them. I think it makes sense to do this, remember the parts most likely to be read are the abstract and the first few lines of the discussion.

Authors' response:

- We have restated the 4 themes in the discussion section as was suggested by the reviewer. (page 25, line 21-24)

The alarming numbers of opioid prescriptions and refill prescriptions among GPs – once again you need to explain this language

Authors' response:

- We partly rewrote the discussion section and removed this sentence.

As GPs we recognize the powerlessness felt when after hours of motivational talks with patients suffering chronic pain return from high quality pain centres with opioid prescriptions without further explanation or communication.

Authors' response:

We rephrased: "As GPs, we recognize the powerlessness felt when hours of motivational talks with our patients followed by referral to high quality pain centers results in patients returning to our care with opioid prescriptions and no further explanation or communication. " (page 26, line 24-28)

non-cancer pain – you've changed your term….

Authors' response:

- We have put chronic between brackets before “non-cancer pain”. Our aim was to explore the attitude of GPs for “acute” and “chronic” non-cancer pain. We have put “non-cancer” before pain where applicable to remain consistent throughout the manuscript.

This review describes how GPs’ as “healers of pain” and “guardians of the community” deliberate the use of opioids in chronic pain. Doesn’t make sneeze in English needs rewrite

Authors’ response:

- We rephrased the sentence: “This review describes the difficulties in opioid prescribing encountered by GPs as both “healers of pain” and “guardians of the community” to “This review demonstrates the difficulties encountered by GPs in treating (chronic) non-cancer pain and refraining from opioid prescription..” (page 28, line 1-2)

A zero-tolerance policy towards pain – not sure what you mean by this statement.

Authors’ response:

- This term is mentioned by Desveaux et al, and is explained in the result section “GPs from Desveaux et al, (31) reported that patients expect chronic pain to reach the zero.”

effective communication strategies not only for GPs but also fine-tuned by GPs are necessary to turn the tide on this opioid crisis. This is new material and it’s in your conclusion. It’s an important issue and needs to be discussed above. You do mention communication with specialists above but this sentence is unclear do you mean with patients with staff with other Gps or with specialists?

Authors’ response:

- We discussed this part in the third paragraph of the discussion section. We partly rewrote this paragraph to clarify this. (page 26, line 30 and onwards)

I think you’re right and that the thing you have to offer that is different is he rethinking from a GP perspective, I’d love to read more about this, so expand on it if you can. Can you say that what would work for you in your setting is the some as GPs from other countries and other systems?

Authors’ response:

- We believe recommendations such as educating GPs and patients, improving collaboration between GPs and specialists, and developing guidelines for GPs fine-tuned by GPs will work across different healthcare systems. Especially guidelines developed by GPs is an important one, because co-designing guidelines with end-users (GPs) will influence their success.
- It is challenging to think about health care system specific solutions. In the Netherlands health care budget for physiotherapy has been cut in past years. Changing health policies such as these, might not solve the whole problem of increased opioid use in the Netherlands but could be a step in the right direction. Recently, a consortium has been installed in the Netherlands to investigate the causes and consequences of opioid use and also aim to influence policy level changes We added these considerations to the discussion section. (page 27, line 4-18)

If this is part of PhD you do need to mention that.

Authors' response:

- Added the following sentences to the funding statement: "Author LK is a PhD student and participated to this work as part of her PhD project."